

Enterprise Aviation Data Platform (EADP)

STAR SCHEMA — DETAILED DESIGN DOCUMENT

PostgreSQL Data Warehouse | 7 Dimensions + 7 Fact Tables

Purpose: This document describes only the EADP dimensional Star Schema: its structure, dimensions, fact-table grains, keys, relationships, and analytical role.

Warehouse Layer	Schema	Model
Gold / Analytical	warehouse	Dimensional Star Schema
Dimensions	7	Conformed descriptive entities
Facts	7	Operational and analytical events

1. Overall Star Schema

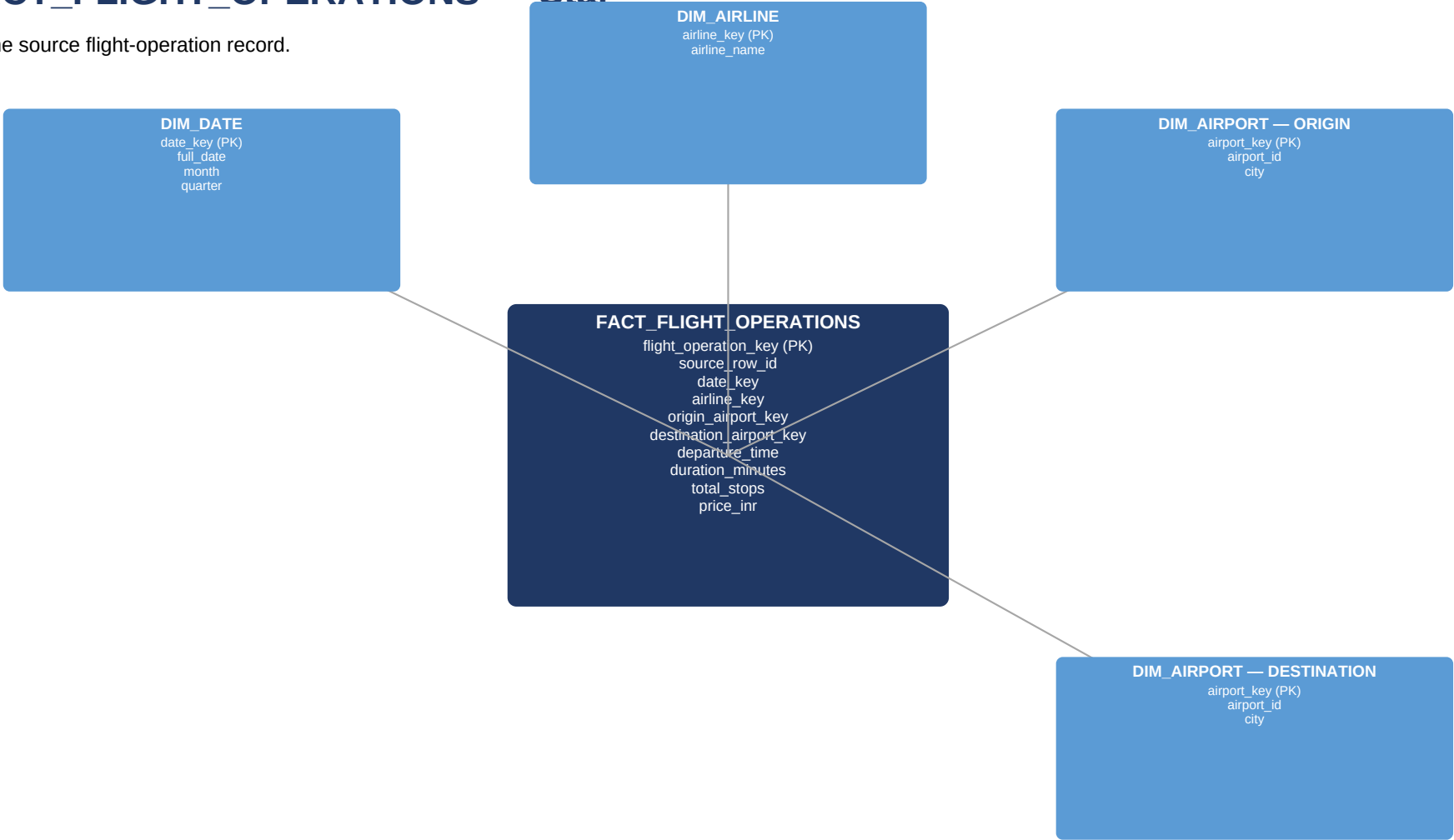
The EADP warehouse uses dimensional modeling. Each fact table represents a specific business process at a defined grain. Dimension tables contain descriptive attributes used to slice and aggregate facts. Surrogate dimension keys provide stable warehouse relationships, while fact tables store foreign keys to the dimensions relevant to each process.

Dimension	Key	Fact relationships
DIM_DATE	date_key	Provides descriptive context to one or more facts
DIM_AIRLINE	airline_key	Provides descriptive context to one or more facts
DIM_AIRPORT	airport_key	Provides descriptive context to one or more facts
DIM_AIRCRAFT	aircraft_key	Provides descriptive context to one or more facts
DIM_PASSENGER	passenger_key	Provides descriptive context to one or more facts
DIM_CREW	crew_key	Provides descriptive context to one or more facts
DIM_FLIGHT	flight_key	Provides descriptive context to one or more facts

Fact processes: Flight Operations • Flight Delay • Booking • Baggage • Maintenance • Crew Scheduling • Aircraft IoT

2. FACT_FLIGHT_OPERATIONS — Star

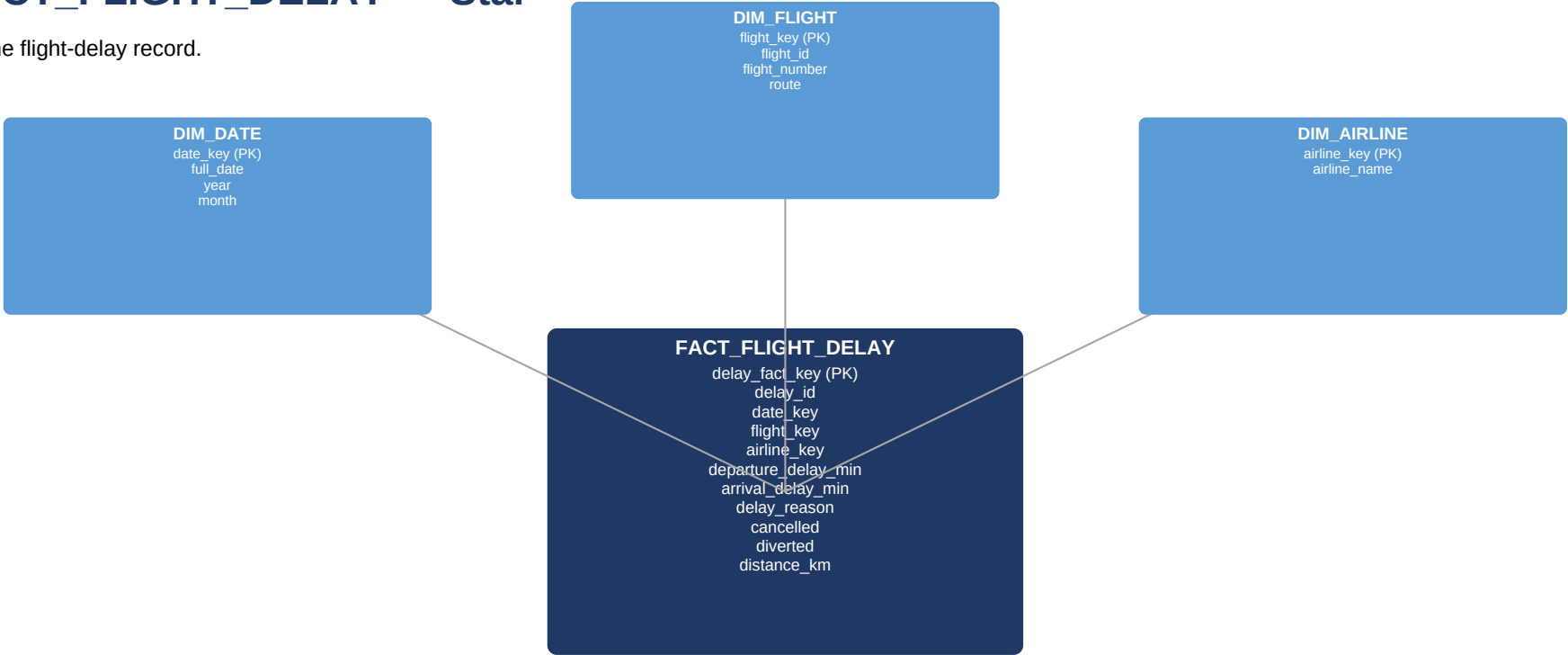
Grain: One source flight-operation record.



Relationships: `date_key` → `dim_date`; `airline_key` → `dim_airline`; `origin_airport_key` and `destination_airport_key` → `dim_airport`.

3. FACT_FLIGHT_DELAY — Star

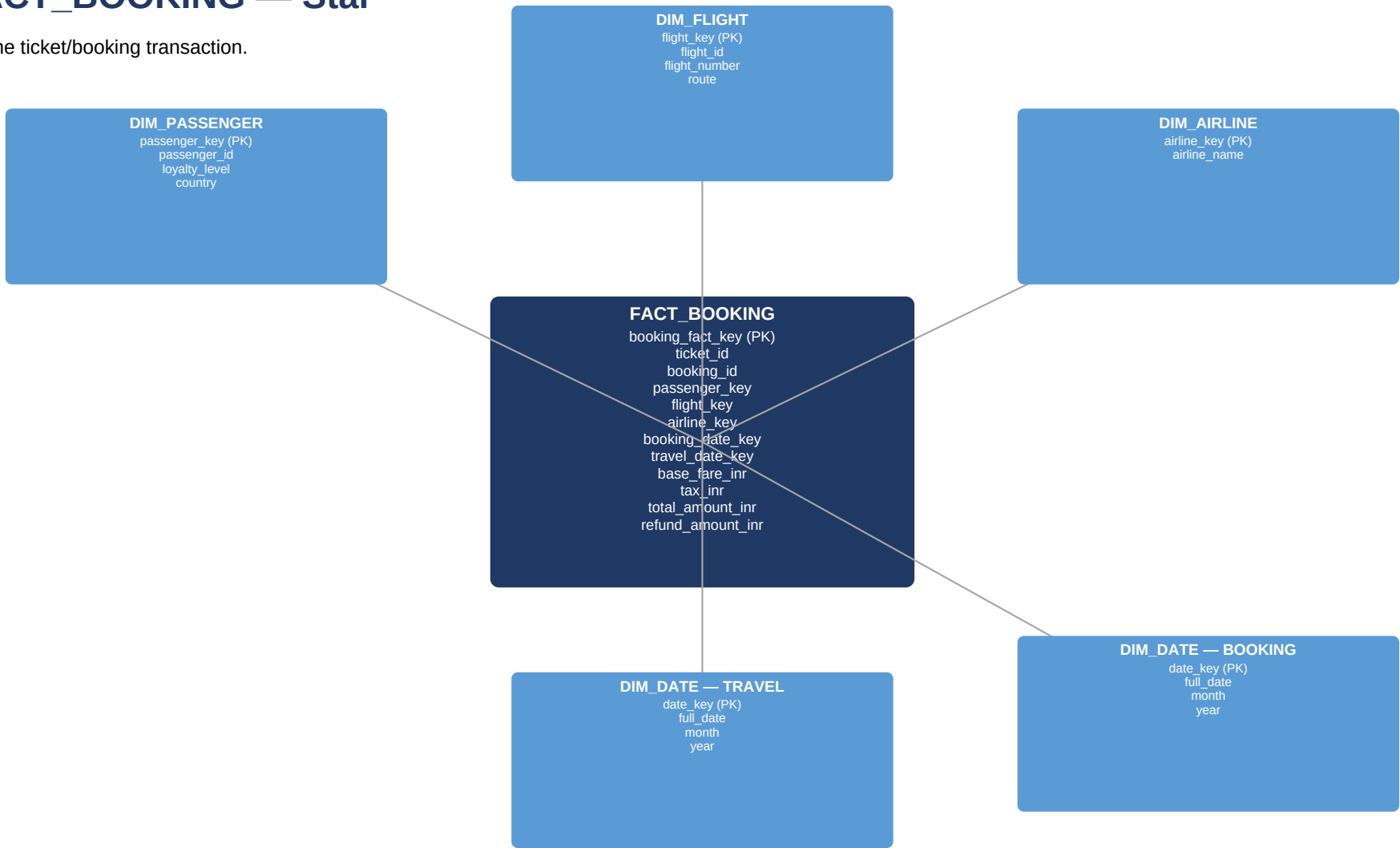
Grain: One flight-delay record.



Relationships: `date_key` → `dim_date`; `flight_key` → `dim_flight`; `airline_key` → `dim_airline`.

4. FACT_BOOKING — Star

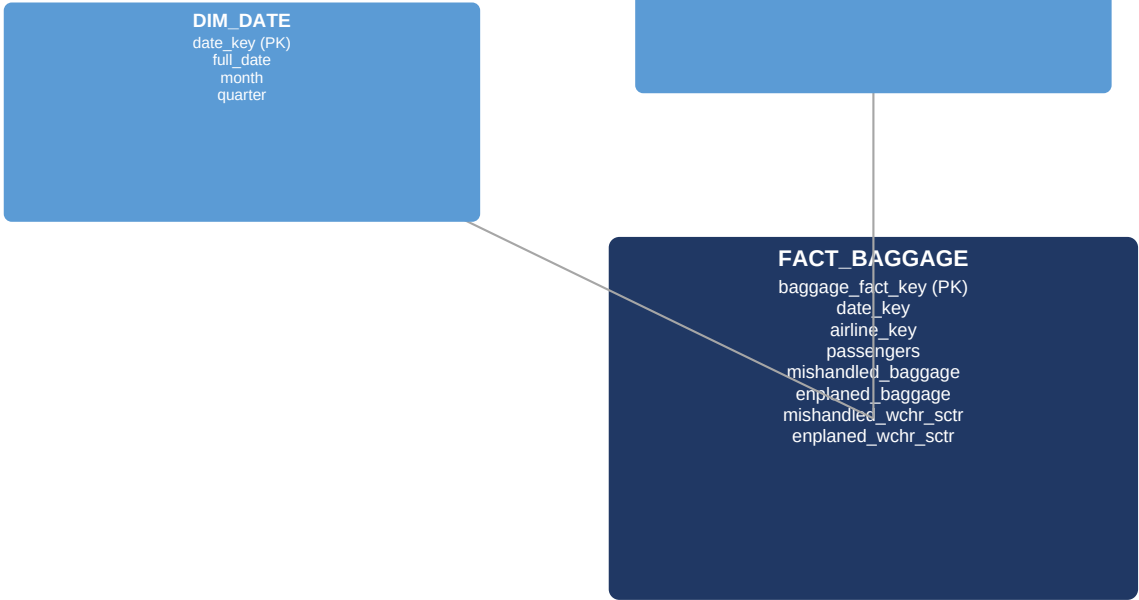
Grain: One ticket/booking transaction.



Relationships: passenger_key → dim_passenger; flight_key → dim_flight; airline_key → dim_airline; booking_date_key/travel_date_key → dim_date.

5. FACT_BAGGAGE — Star

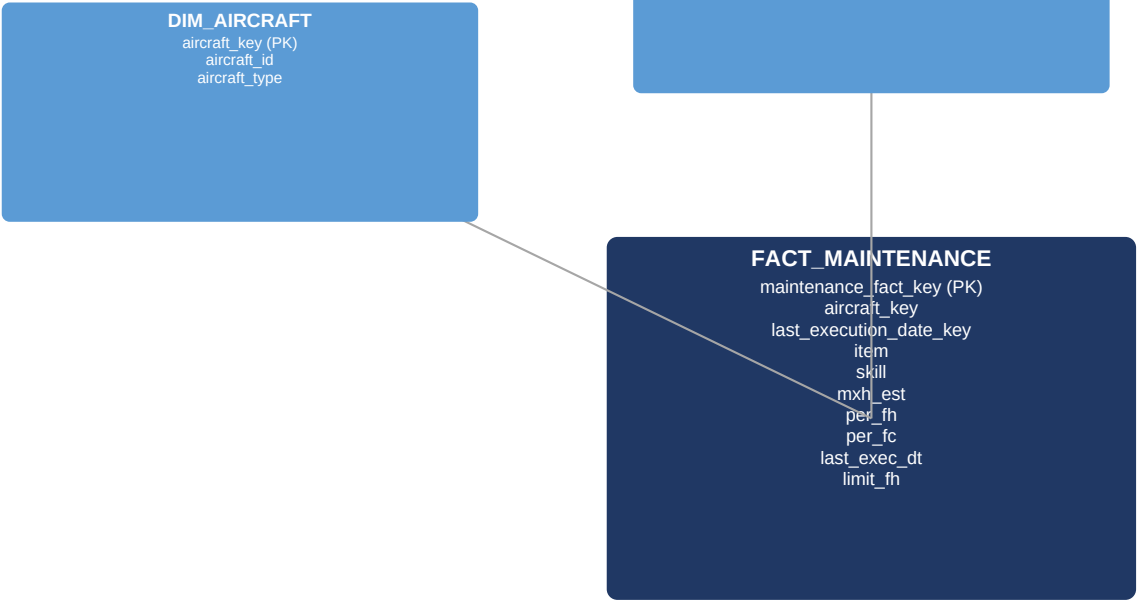
Grain: One airline/month/form-type baggage record.



Relationships: date_key → dim_date; airline_key → dim_airline.

6. FACT_MAINTENANCE — Star

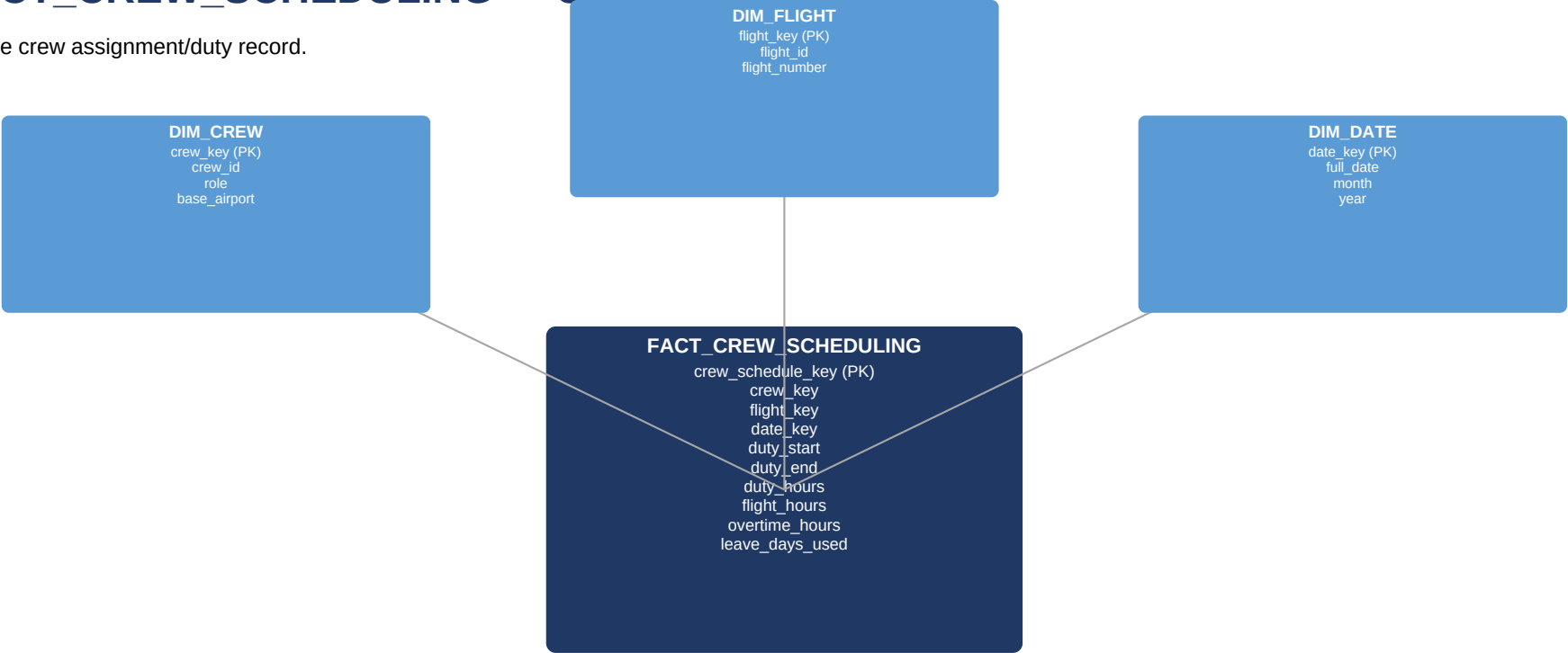
Grain: One maintenance-task record for an aircraft.



Relationships: aircraft_key → dim_aircraft; last_execution_date_key → dim_date.

7. FACT_CREW_SCHEDULING — Star

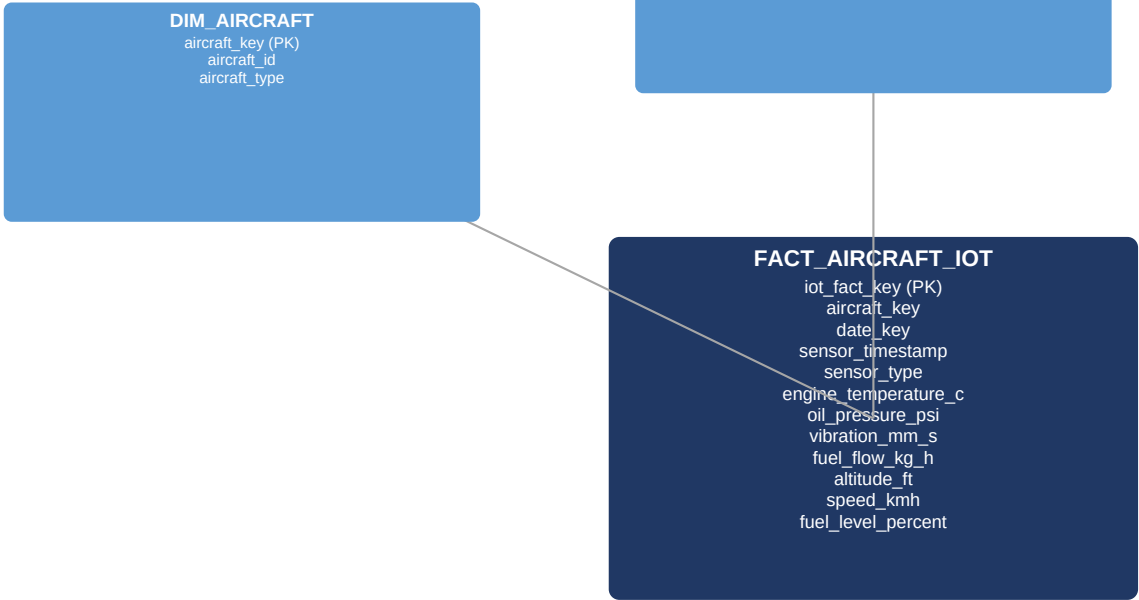
Grain: One crew assignment/duty record.



Relationships: `crew_key` → `dim_crew`; `flight_key` → `dim_flight`; `date_key` → `dim_date`.

8. FACT_AIRCRAFT_IOT — Star

Grain: One aircraft sensor observation.



Relationships: aircraft_key → dim_aircraft; date_key → dim_date.

9. Dimension Reference

Dimension	Primary Key	Important Attributes	Role
dim_date	date_key	full_date, day, month, month_name, quarter, year, week_of_year, day_of_week, day_name	Calendar / time analysis
dim_airline	airline_key	airline_name, carrier_code, unique_carrier, unique_carrier_name	Airline analysis
dim_airport	airport_key	airport_id, airport_name, city, state, country, ICAO, latitude, longitude, elevation, status	Airport / route analysis
dim_aircraft	aircraft_key	aircraft_id, aircraft_type, airline	Aircraft analysis
dim_passenger	passenger_key	passenger_id, customer_id, name, age, gender, loyalty, spend, contact attributes	Passenger / customer analysis
dim_crew	crew_key	crew_id, crew_name, role, airline, base_airport	Crew analysis
dim_flight	flight_key	flight_id, flight_number, airline_key, origin_airport_key, destination_airport_key	Flight / route analysis

10. Fact–Dimension Relationship Matrix

Fact Table	Date	Airline	Airport	Aircraft	Passenger	Crew	Flight
fact_flight_operations	✓	✓	✓	—	—	—	—
fact_flight_delay	✓	✓	—	—	—	—	✓
fact_booking	✓*	✓	—	—	✓	—	✓
fact_baggage	✓	✓	—	—	—	—	—
fact_maintenance	✓	—	—	✓	—	—	—
fact_crew_scheduling	✓	—	—	—	—	✓	✓
fact_aircraft_iot	✓	—	—	✓	—	—	—

* `fact\_booking` has two role-playing references to `dim\_date`: `booking\_date\_key` and `travel\_date\_key`.

11. Star Schema Design Principles Used

Defined grain — Every fact table has an explicit business grain so measures can be interpreted correctly.

Surrogate dimension keys — Dimensions use warehouse surrogate keys such as `airline_key`, `airport_key` and `flight_key`.

Foreign-key relationships — Facts reference the dimensions needed for analysis.

Conformed Date dimension — `dim_date` is shared across processes, with role-playing keys where multiple dates exist in one fact.

Degenerate/source identifiers — Business/source identifiers such as `ticket_id`, `booking_id` and `delay_id` remain in facts where useful.

Role-playing dimensions — `dim_airport` is used for both origin and destination; `dim_date` is used for booking and travel dates.

Analytical separation — Descriptive attributes are stored in dimensions while event measures and transactional attributes are stored in facts.

12. EADP Star Schema Summary

The final warehouse consists of seven dimensions and seven fact tables. The dimensions provide reusable analytical context, while the facts capture flight operations, delays, bookings, baggage, maintenance, crew scheduling and aircraft IoT observations. This structure supports SQL reporting and downstream BI analysis while preserving clear business grains and foreign-key relationships.